Real Estate Data Analysis Report

Prepared for the management of the real estate company

Important: all data in this report is **synthetic** (randomly generated with NumPy, seed 42). It does not describe real properties or the real Pakistani property market.

1. Data overview and data quality

The dataset has 500 property records and 15 columns across five cities. 113 missing cells (bedrooms, bathrooms, price, parking, distance, locality and owner type) were filled using medians or 'Unknown', and 5 duplicate listings were removed. **495 properties** were analysed.

2. Property prices (PKR)

Statistic	Price (PKR)
Average	27,018,889
Median	16,000,000
Minimum	1,450,000
Maximum	209,850,000

Most expensive city (average price): **Islamabad**. Most affordable city: **Faisalabad**. Most common property type: **House** (47.5% of listings).

3. City and property-type comparisons

City	Listings	Avg price (PKR M)	Median size (Marla)	Avg price/Marla (PKR M)
Islamabad	96	38.6	6.0	4.06
Lahore	139	31.6	6.0	3.65
Karachi	131	26.3	5.0	2.90
Multan	67	14.3	7.0	1.32
Faisalabad	62	13.9	6.5	1.56

Average price by city and property type (PKR millions):

City	Commercial	Flat	House	Plot
Faisalabad	18.0	6.1	14.4	19.4
Islamabad	29.4	14.6	42.6	79.5
Karachi	46.9	9.0	26.0	35.1
Lahore	47.2	9.7	39.1	31.8
Multan	13.5	4.0	19.5	16.8

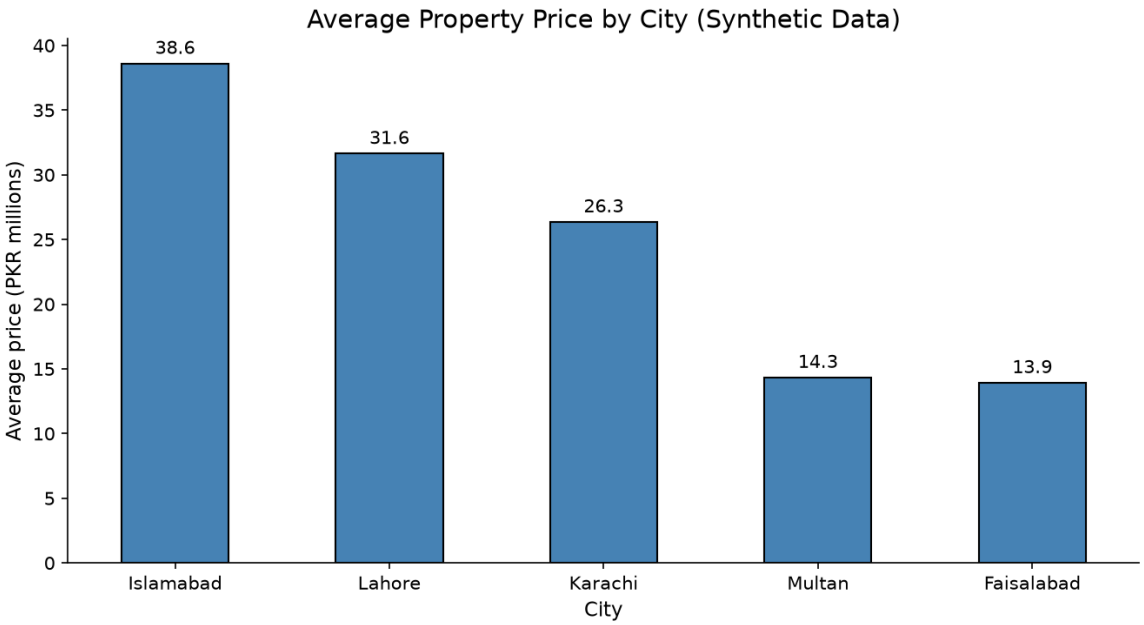


Figure 1: Average property price by city.

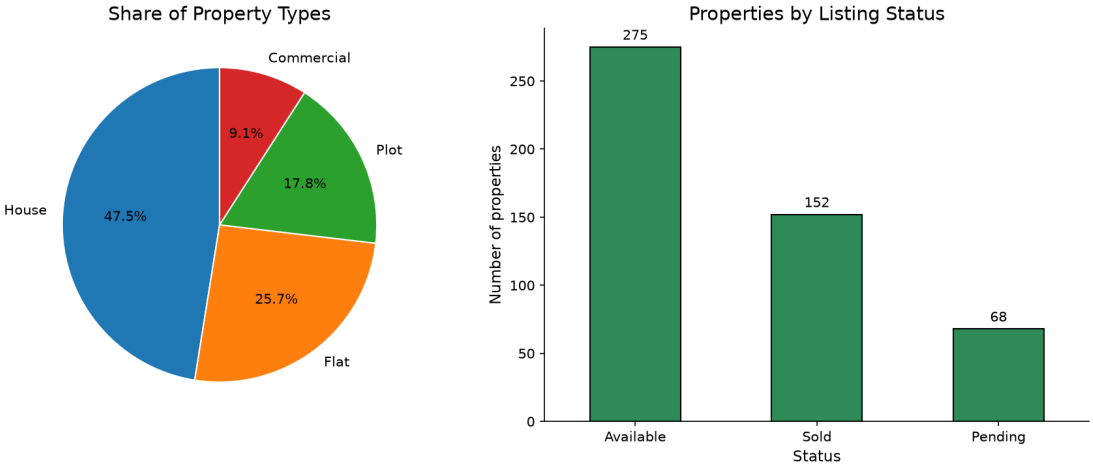


Figure 2: Share of property types and listing status.

4. Distributions, relationships and monthly trends

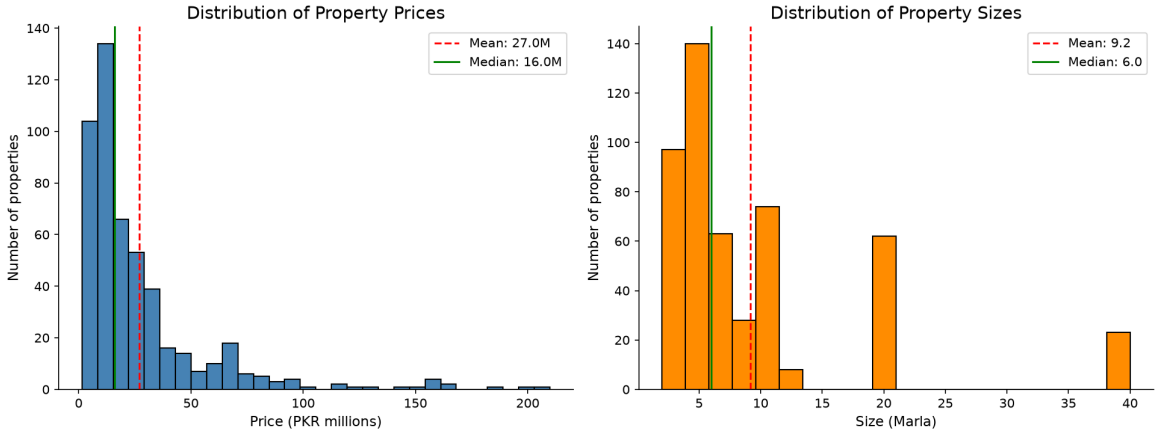


Figure 3: Distribution of prices and sizes (right-skewed).

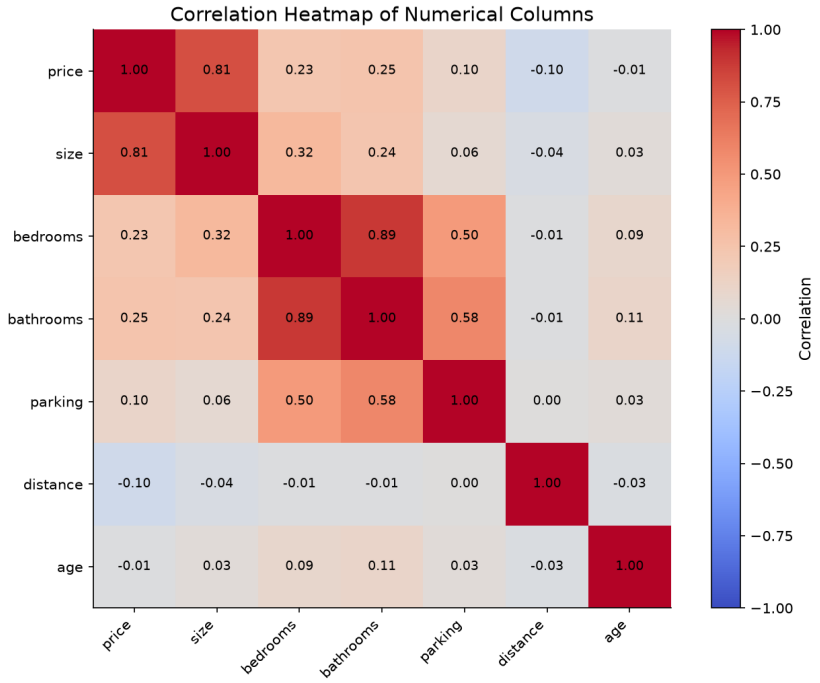


Figure 4: Correlation heatmap; size is the strongest price driver.

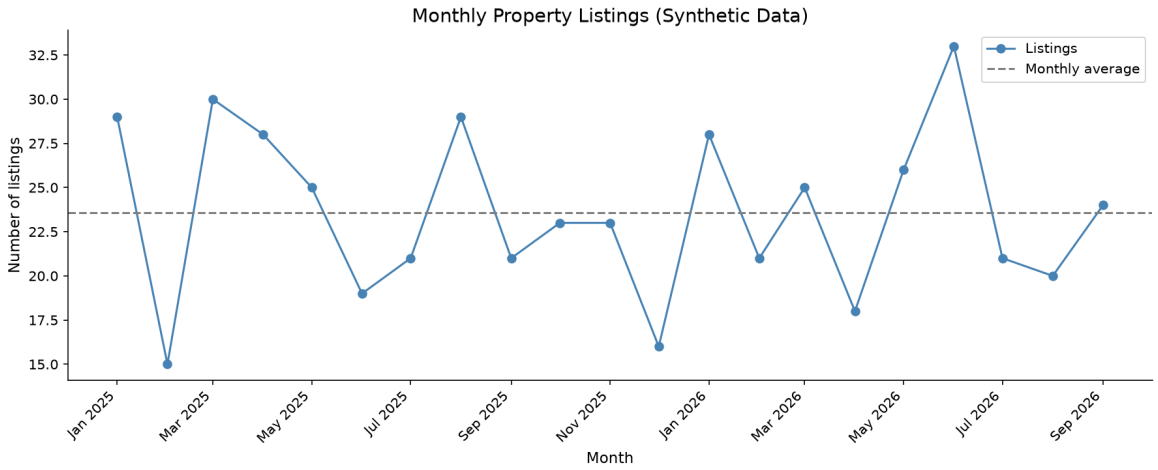


Figure 5: Monthly listings (no clear trend).

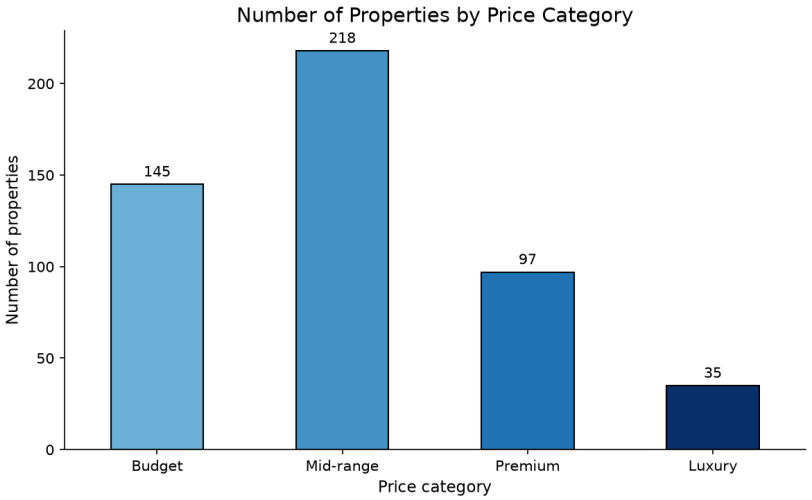


Figure 6: Number of properties by price category.

5. Business insights (calculated from the data)

1. Data quality: 500 records were loaded; 113 missing cells were filled and 5 duplicate listings were removed, leaving 495 properties for analysis.

2. Prices are strongly right-skewed: the average price (PKR 27.0M) is well above the median (PKR 16.0M) because a few very expensive properties pull the mean up.

3. Islamabad has the highest average property price (PKR 38.6M) and Faisalabad the lowest (PKR 13.9M). By price per Marla, Islamabad is the most expensive city (PKR 4.06M per Marla).

4. House is the most common property type (47.5% of listings), while Commercial properties have the highest median price per Marla (PKR 5.38M).

5. Size is the main price driver: the correlation between size and price is 0.81, while distance to the city centre (-0.10) is only weakly related to total price.

6. 55.6% of properties are Available, 30.7% Sold and 13.7% Pending.

7. 44.0% of properties are Mid-range (PKR 10M to 30M), compared with 29.3% Budget, 19.6% Premium and 7.1% Luxury.

8. Monthly listings range from 15 to 33 (busiest month: June 2026) with no clear upward or downward trend (-0.05 listings per month).

6. Recommendations

1. Price and compare properties using price per Marla within each city and property type, not total price alone, because total price mainly reflects property size.

2. Concentrate pricing and marketing effort on the Mid-range segment (44.0% of listings), and treat the small Luxury segment with individual valuations because it contains the extreme prices.

3. Improve data capture: make bedrooms, bathrooms, price and locality required fields and check new listings for duplicates, so fewer values need to be estimated.

Note: because the data is synthetic, these insights and recommendations illustrate the analysis method; they should be re-checked with real market data before any business decision.